Introduction:

Advanced Driver Assistance Systems (ADAS) is a broad term used for smart safety/assistance features that are equipped in modern cars. Lane Keeping Assist LKA systems are also included in this category. An LKA system is a safety feature that helps a vehicle in preventing unintentional lane departures. Other examples of ADAS include Automatic Emergency Braking (AEB), Adaptive Cruise Control (ACC), and Blind Spot Detection etc.

A. Criteria for Choosing an LKA System

When we started, we had to figure out what a "good" Lane-Keeping Assist system actually needs to work in a place like Pakistan. We broke it down into four main criteria's:

1.) Sensor Technology:

These are the eyes of a vehicle that enable it to see and perceive the environment. For the LKA, we identified three sensor options namely cameras, LIDARs and Sensor Fusion.

- **Camera:** This is the most common and budget friendly option available for LKA systems. But we decided to target the Pakistani market, and this might not have been the best option. This is due to the fact that the road marking are often faded, poorly marked lanes, presence of dust and glare etc.
- **LIDAR:** This is another option used for an LKA system. Although it is very expensive but has very accurate results. It uses laser beams to map the surroundings and can detect physical road edges and curbs even when paint is missing, making it much more reliable for our local context.
- **Sensor Fusion:** This option is seen in very high end autonomous vehicles. The presence of camera and LIDAR can enable us to get the color recognition benefits of the camera and the precision of a LIDAR. In such cases, the camera acts as the primary eye when lanes are visible, but the LiDAR takes over as a "fail-safe" when markings are faded or the weather is poor.
- **Decision:** For now, we have decided to go with a LIDAR such as the Velodyne VLP-16 also known as Puck for the future simulation. This is a widely used LIDAR in autonomous vehicle research. It has a 360 deg field

of view and a detection range of 100m. The datasheet for this LIDAR is also available publically.

2.) Control Algorithm:

This is the brain of the system that tells it how much to steer and how to steer. We also evaluated three options for this.

- **PID:** This is the "classic" approach we've used in NERC. It reacts to where the car is *right now* by looking at current error i.e. how far the car is from center of lane and calculates steering correction. While it is good at making very fast, small corrections to keep the car steady right now but it struggles with curved roads.
- **MPC:** Model Predictive Control is a bit more complex algorithm used for LKA systems. It looks ahead at the road and plans the steering for the next few seconds in advance which makes steering through curves much smoother.
- **Hybrid:** A hybrid approach can also be adopted by integrating both algorithms. In such case, we can chose to use MPC to calculate the "target" steering angle for the path ahead, and a PID loop to handle the high-speed execution of that steering command. This makes the steering feel natural and prevents the shaking behavior or sudden turns.
- **Decision:** For now, we have decided to proceed with MPC control algorithm as it is far better for LKA. It handles road curves much more smoothly than a PID controller because it proactively plans the steering sequence rather than just reacting to an error.

3.) Vehicle Dynamics:

This describes the reaction of the car to the command send by the control algorithm i.e. how the car reacts physically to execute that steering command. We considered two models for this.

- **Dynamic Bicycle Model:** This model is very complex and includes tire slip, air resistance, and weight shifting. While it is realistic for high-speed scenarios, it requires many parameters like tire-road friction coefficients that are hard to find.

- **Kinematic Bicycle Model:** This treats the car like a bicycle (two wheels) and it focuses purely on the geometry of motion (speed, wheelbase, and steering angle). This is suitable for slow and moderate speeds and is simpler to work with mathematically.
- **Kinematic Car Model:** This approach treats the vehicle as having 4 wheels and is more complex than kinematic bicycle model. Although it is more complex, but its accuracy is not significantly greater than the Kinematic bicycle model because at moderate speeds, the two front wheels behave almost identically.
- **Decision:** For now, we have chosen the Kinematic bicycle model because LKA usually operates at steady highway speeds where complex tire physics aren't as relevant. It provides a perfect balance. It's accurate enough for our simulation without needing data that is impossible for us to measure.

4.) Technical Specifications:

This was a key criteria as the availability of accurate and verified technical data is necessary for accurate modelling of the LKA system. **B. Selection Process and Considerations:**

The selection from the above criteria was such that we needed to make sure that every chosen criteria worked together.

- **Logical Steps:** We started with the problem of the Pakistani roads, which led us to choose LiDAR. Since LiDAR gives us predictive data, MPC was the natural choice for the controller. Then, we picked a car sold locally that fits this setup.
- **The Reference Vehicle:** We evaluated several cars like the MG HS and Changan Oshan X7, but the Toyota Corolla Cross (1.8V HEV variant) was our final choice. This was because our main goal was to find a car that actually has the hardware for steering assist and for which we can find reliable technical data. The Toyota Corolla Cross is pre-equipped with a Lane Tracing Assist and a Steering Assist system ensuring that it has the electronic steering architecture for our MPC controller. Another reason was that we needed a well-documented car to build an accurate Kinematic Bicycle model. Toyota provides clear, official spec sheets that make our math much more accurate.

- **Safety Standards:** Even though this is a simulation, we want it to be safe. We are following ISO 11270, which is an international standard for LKA systems. This ensures that our steering isn't too jerky or dangerous for a driver. This is obtained by setting constraints in the MPC control algorithm like the max steering angle of 35deg obtained by available turning radius of 5.2m for the Toyota Cross.

C. Ensuring Availability of Technical Specifications:

We ensured that the required parameters or important values required for the model are available publicly.

- For the Kinematic Bicycle Model, the required parameters like vehicle mass, wheelbase length and the maximum steering angle for the vehicle are publicly available.
- Similarly, the parameters for the MPC like Sampling time, Prediction horizon are also available. Same is the case with the LIDAR whose detailed specs are available on the datasheet.